Supplementary Material

Urban Land-Use Efficiency

A Stable Metric for SDG 11.3.1 and Its Economic and Structural Drivers: 400 German Districts x 1990-2020 Panel Data Analysis

Sub-national Case Study: Germany (Kreis Level)

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Abstract

This supplement replicates the paper's driver analysis for the Built-up per Capita Rate (BpCR) at the sub-national level, using Germany (district / *Kreis* level) as a case study, and brings in internal (within-country) migration, which the national analysis cannot observe. It asks whether the within-country income-densification and path-dependence results survive at a finer spatial scale.

Keywords: SDG 11.3.1, Built-up per capita rate (BpCR), Sub-national analysis, Germany / Kreise, Internal migration

1. Data and Method

The dependent variable is built exactly as in the main paper, but for German districts (*Kreise*) rather than countries. Urban built-up area V and urban population P come from the GHSL grids (Pesaresi et al., 2024; Schiavina et al., 2023) (urban-masked at GHS-SMOD ≥ 21), aggregated to the GAUL level-2 units for Germany at the five-year epochs 1985–2020; BpCR and urban density are then computed from these totals just as nationally. Three series are taken from the German regional database INKAR (BBSR, 2024), published by the Federal Institute for Research on Building, Urban Affairs and Spatial Development (BBSR) and accessed through the author's `inkaR` R package (Çoban, 2026): GDP per capita (workplace-based), the **internal net-migration rate** (*Binnenwanderungssaldo*, reported per 1,000 inhabitants and rescaled here to a percentage of population for comparability with the national control), the latter being the within-country

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[1]The author used the Claude Opus 4.8 and Gemini 3.5 large language models for improving the writing and for coding assistance in the preparation of this supplement. All analysis, results, and conclusions are the author's own, and the author takes full responsibility for the content.

flow that the national UN measure omits, and total district population, which serves as the denominator of the urban population share (GHSL urban population over INKAR total district population). GAUL districts are matched to INKAR *Kreise* on a cleaned base name, disambiguating same-named *kreisfreie Stadt* / *Landkreis* twins by geographic area, so that 399 of Germany's 400 districts enter the regressions. The single exception is Freyung-Grafenau, a rural Bavarian-Forest district whose largest settlement reaches only the rural-cluster class (GHS-SMOD 13): no cell attains the urban threshold (SMOD ≥ 21), so its urban built-up area and population are zero and BpCR is undefined. This is the same urban definition used throughout the paper, under which a unit with no urban cells is simply not an urban observation, not a matching failure. The panel spans the epochs 1995–2020 for which the INKAR controls are available (2394 district-periods). All specifications carry district and period fixed effects with standard errors clustered by district; the dynamic models add the lagged dependent variable and use Arellano-Bond and Blundell-Bond GMM, as in the main paper.

Figure 1 shows the same SDG 11.3.1 construction at the German district scale, for Berlin and the surrounding Brandenburg countryside. As in the national figure, built-up surface (panel b) is retained only where the GHS-SMOD grid classifies a cell as urban (SMOD at least 21, panel c), giving the urban-masked built-up that enters the analysis as V (panel d); urban population P is masked the same way. The cyan line is the Berlin state boundary: the dense urban core falls inside it, while the rural matrix that the mask removes makes up most of the surrounding districts.

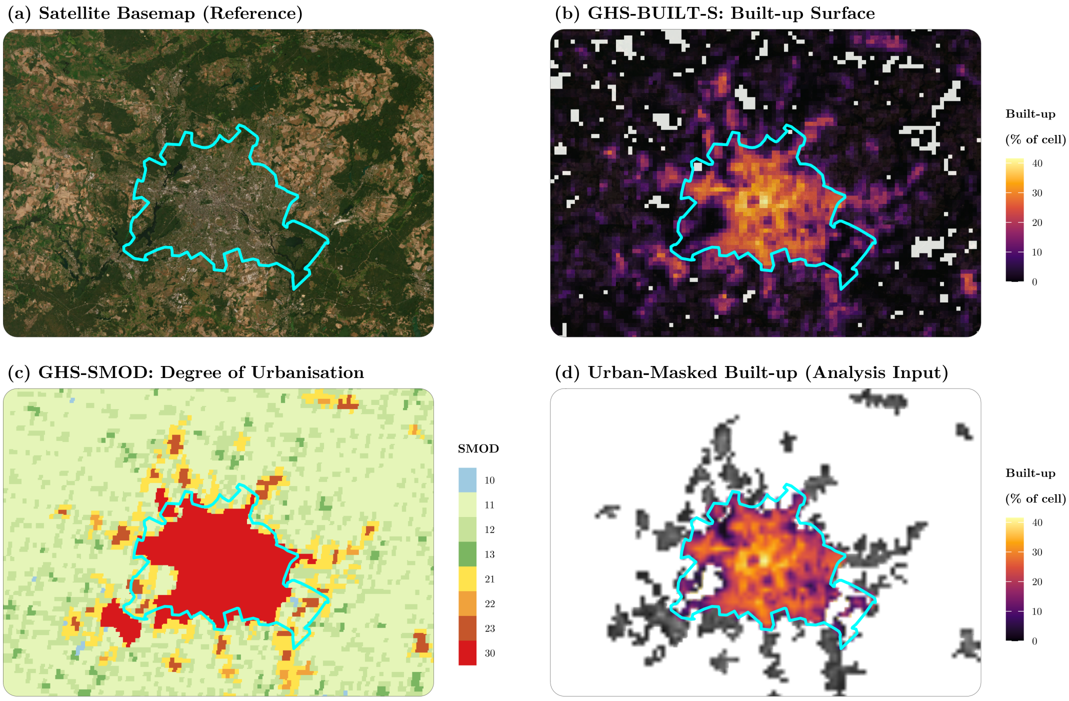

Figure 1: Building the analysis input following the SDG 11.3.1 metadata at the German *Kreis* scale (Berlin / Brandenburg window), each layer drawn over an Esri satellite basemap. (a) Basemap with the Berlin state boundary (cyan). (b) GHS-BUILT-S built-up surface, shown as the built-up share of each 1 km cell. (c) GHS-SMOD Degree of Urbanisation (1 km); cells coded 21 or higher count as urban. (d) Urban-masked built-up, the input actually used: panel (b) kept only where panel (c) is urban.

2. Drivers of Per-Capita Land Use

We estimate the drivers of BpCR at the district level exactly as in the main paper: a two-way fixed-effects model built up control-by-control, then Arellano-Bond (difference) and Blundell-Bond (system) GMM that add the lagged dependent variable to capture path-dependence and correct its Nickell bias. The key addition relative to the national models is **internal (within-country) net migration**, the INKAR *Binnenwanderungssaldo*, which the national panel (international migration only) cannot observe.

Table 1: Dynamic-panel model at the German Kreis level: FE stepwise build-up, Arellano-Bond (difference) and Blundell-Bond (system) GMM.

	Dependent variable: BpCR						
	Fixed Effects					GMM (Bias-Corrected)	
Variables	Base	+ Density	+ Net Migr.	+ GDP	+ Urban %	Arellano-Bond	Blundell-Bond
BpCR[t−1]	+0.2487***	+0.2266***	+0.2240***	+0.2219***	+0.2191***	+0.7085***	+0.8497***
	(0.0424)	(0.0457)	(0.0445)	(0.0446)	(0.0435)	(0.1221)	(0.0566)
ln(Urban Density)		-0.0079	-0.0083	-0.0091	-0.0080	-0.0443***	-0.0003
		(0.0056)	(0.0056)	(0.0058)	(0.0057)	(0.0054)	(0.0007)
Internal Net Migration (% Pop)			-0.0003*	-0.0003*	-0.0003*	-0.0001	-0.0001
			(0.0002)	(0.0002)	(0.0002)	(0.0001)	(0.0002)
ln(GDP per capita)				-0.0019	-0.0023	-0.0021	-0.0004
				(0.0016)	(0.0017)	(0.0018)	(0.0005)
Urban Pop. Share					-0.0112	-0.0263**	-0.0017***
					(0.0109)	(0.0107)	(0.0006)
Observations	2,394	2,394	2,394	2,394	2,394	1,995	2,394
Within R^2	0.065	0.070	0.074	0.075	0.078	–	–
District FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Period FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Instruments	–	–	–	–	–	11	17
Hansen (p)	–	–	–	–	–	0.75	0.00
AR(1) (p)	–	–	–	–	–	0.03	0.00
AR(2) (p)	–	–	–	–	–	0.30	0.01

Note: *** p<0.01, ** p<0.05, * p<0.1. Standard errors in parentheses: clustered by district in the fixed-effects columns; two-step GMM with the Windmeijer finite-sample correction (Windmeijer, 2005) in the Arellano-Bond and Blundell-Bond columns. The lagged dependent variable is instrumented by lags 2 to 3 of BpCR, with the instrument matrix collapsed. In the GMM columns the Observations row reports post-transformation equation observations (the differenced-equation set for Arellano-Bond; the larger level-equation set for Blundell-Bond). Sample: 399 of Germany's 400 districts; Freyung-Grafenau is excluded because no cell reaches the urban threshold (SMOD of at least 21), leaving its urban built-up and population at zero and BpCR undefined.

Three points emerge. First, per-capita land use is **strongly path-dependent** at the district level too: the lagged dependent variable is large and significant ($\rho \approx 0.71$), echoing the persistence found across countries. Second, the **income** coefficient is negative in every specification, matching the sign of the national income-densification result, but it does not reach statistical significance in this shorter panel, so the district evidence is consistent with the national finding rather than an independent confirmation of it. Third, **internal net migration** enters with an essentially zero coefficient that is only marginally significant (10% level) in the fixed-effects columns and insignificant in the GMM: six five-year epochs give little power to identify a within-district migration channel. Urban density, insignificant in the fixed-effects columns, turns significantly negative in the Arellano-Bond column (-0.044), mirroring the national compact-city estimate; as in the main paper, this coefficient shares the population term with the dependent variable and is read as partly mechanical rather than as clean evidence.

3. Limitations

Several limitations qualify this case study and mark it as a corroborating local illustration rather than a stand-alone causal analysis.

Coverage and matching. GAUL districts are matched to INKAR by name (399 of Germany's 400 districts); the only exclusion, Freyung-Grafenau, has no urban-classified cells and hence no defined BpCR. The match relies on a cleaned base name with an area-based rule for same-named city/rural twins rather than an official code, so it is accurate here but would need a code crosswalk to be fully automatic in other settings.

Short, recent panel. The INKAR controls are available only from 1995, so the estimation window is six five-year epochs (1995–2020). This is shorter than the national panel and, together with the five-year spacing, means the models speak to medium-run trajectories rather than short-run adjustment. The brevity matters most for the dynamic estimators: because the GHSL series used here begins in 1985 (BpCR from 1990), the deeper lags that instrument the Arellano-Bond difference equation are available only from 2000 onward, so the GMM is estimated on fewer periods than the within (fixed-effects) columns even though the latter retain all of 1995–2020.

GMM diagnostics. The Arellano-Bond difference estimator is well behaved: the Hansen test does not reject the instruments, first-order serial correlation is present and second-order is absent, as required. The Blundell-Bond system estimator, by contrast, is rejected at this fine and short scale (its Hansen and AR(2) tests fail), so we treat Arellano-Bond as the preferred dynamic estimator and read Blundell-Bond only as a sign-and-significance cross-check.

Internal migration. The internal net-migration control (INKAR *Binnenwanderungssaldo*) is the within-country flow the national analysis cannot observe, but within districts its coefficient is essentially zero, at best marginally significant (10% level) in the fixed-effects columns and insignificant in the GMM. A plausible reading is that any clearer migration-densification association is cross-sectional (denser districts attract internal migrants), a pattern this within-district design does not test formally; six epochs in any case give little power to identify such a dynamic.

Measurement and scope. As in the main paper, the built-up and population layers are satellite-based estimates, the urban area is defined at a fixed Degree-of-Urbanisation threshold (GHS-SMOD of at least 21), and present-day GAUL boundaries are applied throughout; the income and migration controls are administrative aggregates. The estimates are associations, not causal effects, and they describe a single country, so they illustrate rather than establish external validity.

4. Conclusion

This supplement takes the paper's driver analysis to the sub-national (German district) scale, across 2394 district-periods. The headline replication is path-dependence: per-capita urban land use is strongly persistent within districts (Arellano-Bond $\rho \approx 0.71$), echoing the finding across countries, so this result is not an artefact of country-level aggregation. The income coefficient is negative throughout, in line with the sign of the national income-densification result, but it remains statistically insignificant in this short six-epoch panel; the district evidence therefore corroborates the direction of the national finding without independently establishing it.

The case study also adds the internal migration that the national analysis cannot see. The within-country net-migration flow enters with an essentially zero coefficient, at best marginally significant in the fixed-effects columns and insignificant in the GMM; identifying that channel convincingly would require a longer panel. Taken together, the German evidence confirms the strong persistence of per-capita land use at a scale the national panel cannot reach, and is directionally consistent with, though not conclusive on, the income-densification channel.

References

BBSR, 2024. INKAR: Indikatoren und karten zur raum- und stadtentwicklung. Federal Institute for Research on Building, Urban Affairs and Spatial Development (BBSR), Bonn. URL: https://www.inkar.de/. regional indicators database; GDP per capita and internal net-migration rate at the district (Kreis) level.

Çoban, Ö.F., 2026. inkaR: Download and Analyze Spatial Development Data from INKAR. URL: https://CRAN.R-project.org/package=inkaR, DOI:10.32614/CRAN.package.inkaR. R package version 0.6.2.

Pesaresi, M., Schiavina, M., Politis, P., Freire, S., Krasnodębska, K., Uhl, J.H., Kemper, T., et al., 2024. Advances on the Global Human Settlement Layer by joint assessment of Earth Observation and population survey data. International Journal of Digital Earth 17. DOI:10.1080/17538947.2024.2390454.

Schiavina, M., Melchiorri, M., Pesaresi, M., 2023. GHS-SMOD R2023A - GHS settlement layers, application of the Degree of Urbanisation methodology (stage i) to GHS-POP R2023A and GHS-BUILT-S R2023A, multitemporal (1975-2030). URL: http://data.europa.eu/89h/a0df7a6f-49de-46ea-9bde-563437a6e2ba, DOI:10.2905/A0DF7A6F-49DE-46EA-9BDE-563437A6E2BA.

Windmeijer, F., 2005. A finite sample correction for the variance of linear efficient two-step gmm estimators. Journal of Econometrics 126, 25–51. DOI:10.1016/j.jeconom.2004.02.005.